Codetopia Mentorship Program - 2026

Git & GitHub

Session 5 : Remote Repositories

What we will cover today

- Local vs Remote — why you need both
- Creating a GitHub account
- Setting up SSH authentication (Ed25519)
- Testing your connection to GitHub
- Session Project: Secure Connection Certificate

Quick Review From Previous Session

- You need to inspect what changed in a specific commit. Which command do you use, and what information does it show you?
- Your teammate accidentally pushed a bad commit to a shared branch. Should you use `git revert` or `git reset` to undo it, and why does the choice matter?
- You staged a file by mistake before committing. What single command restores it to unstaged without losing your edits?
- In your Session 4 project, you created undo-toolkit.txt. Without opening it, what does `git reset --soft HEAD~1` do, and when should you only use it?

Remote Repositories

Local vs Remote

What is remote repository

- Your local repo lives on your computer, only you can see it
- A remote repo (on GitHub) lives in the cloud, shareable and backed up
- GitHub is the most popular hosting platform, it adds Pull Requests (PRs), issues, and teams on top of Git

Local vs Remote Repositories

Local Repos	Remote (GitHub)
On your machine	It is in the cloud
Only you see it	Shareable with anyone
Only you see it	Always accessible
Git commands	Git + GitHub UI

Setting Up SSH Authentication

Why SSH instead of HTTPS?

- SSH uses a key pair — no typing passwords every time you push
- More secure than password authentication
- Industry standard for developer authentication

Generate your SSH key (Ed25519 — recommended)

```
ssh-keygen -t ed25519 -C "you@example.com"
```

Copy the public key and add to GitHub

```
cat ~/.ssh/id_ed25519.pub
```

Setting Up SSH Authentication

Copy the public key output → Copy the output → GitHub → Settings → SSH Keys → New SSH Key → Paste and save

Test the connection

```
ssh -T git@github.com
#Expected Output
Hi username! You've successfully authenticated.
```

Session Project

Secure Connection Certificate

Session Project

Your deliverables

1. Generate an SSH key pair (Ed25519) and add the public key to GitHub and take a screenshot of it.
2. Run ssh -T git@github.com and screenshot the success message
3. Create an empty repo on GitHub with a README, and copy the SSH clone URL and clone it locally.
4. Write a 5-line explanation of why SSH is safer than HTTPS passwords in the README file

References & Further Reading

Resources

Text Resource

- The Odin Project: Introduction to Github
  https://www.theodinproject.com/lessons/foundations-introduction-to-github

Video Resource

Traversy Media — Git & GitHub Crash Course
Git & GitHub Crash Course For Beginners